

TIM ZHANG

Data Engineer | AI-Driven Pipeline Architect | Data + AI Systems Builder
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PROFESSIONAL SUMMARY

Data Engineer with 8+ years of enterprise data warehouse, cloud migration, and production data pipeline experience, now focused on AI-driven data systems, Agentic RAG, financial risk systems, and clinical AI applications. Delivered 10TB+ cloud migration, 5-layer EDW architectures, and SparkSQL optimization with 87.5% processing improvement. Recent flagship work includes an award-winning Lingnan Cup SRR Agentic Case Processing System, MCC FWA insurance fraud graph intelligence system, and GaitGPT clinical gait analysis agent. Portfolio and public showcase repositories provide case-study visuals, demo assets, and reference code.

TECHNICAL SKILLS

Programming & Data: Python, SQL, SparkSQL, Impala SQL, Bash/Shell, Java

Data Engineering: Hadoop, Spark, Hive, StarRocks, Kafka, Seatunnel, DolphinScheduler, Informatica PowerCenter

AI/LLM Systems: FastAPI, LangChain/LangGraph, RAG, Agentic workflows, pgvector, GraphDB/Neo4j, OpenAI GPT-4o, Gemini, LLM-as-Judge

Cloud & DevOps: Google Cloud Run, Cloud SQL, Tencent Cloud, Docker, Linux, GitHub Actions

Domains: Government case automation, insurance FWA, claims adjudication, banking data warehouse, retail/financial data quality

SELECTED AI & DATA PROJECTS

SRR Agentic Case Processing System | *Technical Lead* | 99.4% contribution | Lingnan Cup 2nd Prize | Featured on TV | 2025-2026

Project links: [Portfolio case study](#) | [Public showcase repo](#)

- Designed a 7-layer Agentic processing architecture with 17 pluggable atomic capabilities for Hong Kong ArchSD SRR case automation.
- Built multi-format parsing and A-Q field extraction for ICC 1823, TMO, and RCC channels, with semantic retrieval using PostgreSQL + pgvector and RRF fusion.
- Implemented 3-tier quality assessment, Best-of-N self-healing, and rollback logic to make LLM outputs evaluable and recoverable.
- Delivered 80,430 lines of code across two iterations; 12/12 requirements implemented, with case-study visuals and public reference code available below.

Tech: Python, FastAPI, React/TypeScript, PostgreSQL 15 + pgvector, GPT-4o, Google Cloud Run, Cloud SQL, Docker

MCC FWA: Insurance Claims Fraud Graph Intelligence System | *Core Developer* | Enterprise collaboration | InsurTech / Healthcare AI | 2026

- Built a graph-based insurance FWA pipeline converting 38,659 medical claim JSON files into a property graph with 1.61M nodes and 2.57M edges.
- Designed claim graph schema across Claim, Event, Doctor, Hospital, Diagnosis, Receipt, BillingItem, BreakdownItem, and AmountFeature nodes for traceable risk investigation.
- Developed portable nodes/edges CSV extraction for Neo4j validation and future Google Spanner Graph deployment, with single-claim path inspection for audit explanations.
- Supported AI-assisted claims adjudication with FastAPI async batch processing, LLM-based medical necessity review, policy classification, rejection-code reasoning, and case-level summaries.

Tech: Python, FastAPI, JSON/CSV pipelines, Neo4j, Google Spanner Graph-ready design, LangChain/LangGraph, OpenRouter LLMs

GaitGPT: Literature-Augmented Clinical Gait Analysis Agent | *Architecture Lead* | Manuscript in preparation | iCAN 2026 | LU invention disclosure | 2026

Project links: [Portfolio case study](#) | [Public showcase repo](#)

- Built a privacy-first, self-hosted gait analysis agent connecting quantitative gait parameters, clinical language, and live literature evidence.
- Led project direction and architecture; developed reverse translation agent, knowledge explanation agent, and evaluation framework.
- Implemented bidirectional research-clinical translation, PubMed/Semantic Scholar retrieval, hybrid rule-template-first workflow, and physics-constrained validation.
- Prepared paper/competition/IP package: manuscript in preparation, iCAN 2026 application package, and Lingnan University invention disclosure preparation.
- Evaluated on a 20-question NONSD-Gait benchmark: overall accuracy 0.9223, median response time 10.07s; target venue options include Sensors and medical informatics venues.

Tech: Python, FastAPI, Next.js/React, SurrealDB, LangGraph, PubMed/Semantic Scholar APIs, Ollama/Qwen, OpenAI-compatible providers

PROFESSIONAL EXPERIENCE

Data Engineer | Shenzhen RenruiHR Technology Co., Ltd. | 09/2023 - 09/2025

- Migrated 10TB+ IDC data warehouse to Tencent Cloud and restructured it into a 5-layer ODS/DIM/DWD/DWS/ADS architecture.
- Used Seatunnel for full/incremental synchronization, achieving 99.95% consistency and reducing sync failures by 70%.
- Optimized ORC storage and SparkSQL execution, cutting disk usage by 38%, improving query speed by 55%, and reducing ETL runtime by 87.5%.
- Integrated StarRocks with Hive external tables, moving core analytical queries from 3-10 minutes to avg. 0.8s.

Data Engineer | Hangzhou Yatop Information Technology Co., Ltd. | 04/2021 - 08/2023

- Architected 5-layer Hive data warehouse and reusable model layers, improving cross-team data reuse by 50%.
- Integrated data from 100+ transaction systems with Informatica PowerCenter and built Impala/HQL metrics logic for reporting.
- Created UDFs/UDTFs for complex computation and sensitive data masking, supporting privacy-compliant analytics delivery.

Data Engineer | Shenzhen Owned Technology Co., Ltd. (Client: Bank of Ningbo) | 02/2019 - 03/2021

- Maintained banking data model layers for downstream financial and regulatory reporting.
- Translated data mart requirements into SQL logic and worked directly with banking clients on issue resolution and requirement clarification.

Engineer | Chengdu HQtimes Technology Co., Ltd. | 07/2017 - 11/2018

- Troubleshoot network/system issues for IBM software applications and developed Bash/Python operational automation scripts.

EDUCATION & LANGUAGES

MSc in Artificial Intelligence and Business Analytics | Lingnan University, Hong Kong | 2025 - 2026

Bachelor of Management | Shenzhen University, China | 2014 - 2017

Languages: Cantonese (Native), Mandarin (Native), English (Professional Working Proficiency)